



Bidding Player

Prebid + GAM video header bidding

Publisher Integration Guide

A one-tag drop-in. Runs a Prebid auction against limelightDigital, stitches the winning bid into your GAM ad call, plays the result through video.js + Google IMA.

VERSION

v2.1.0

LICENSE

MIT

REPOSITORY

github.com/aryanvani-projects/bidding-player

ENGINE

cdn.jsdelivr.net/gh/aryanvani-projects/bidding-player@v2.1.0/engine/player.js

DOCUMENTATION

aryanvani-projects.github.io/bidding-player/docs.html

TAG GENERATOR

aryanvani-projects.github.io/bidding-player

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01 — Overview & how it works

One `<script>` tag handles everything: lazy-loads video.js + IMA + Prebid, runs the auction, stitches the full `hb_*` targeting set into your GAM tag's `cust_params`, plays the resulting VAST through IMA.

The engine is hosted at versioned, immutable jsDelivr URLs. Pin to a tag — the bytes cannot silently change.

Auction sequence

- 1 **Engine** parallel-loads video.js, IMA, Prebid; reads `data-*`; registers a video ad unit.
- 2 **Prebid → Limelight**: `POST {host}/ortbh`
- 3 **Limelight** responds with a bid (or no-bid). Winning VAST cached at Prebid Cache; UUID returned.
- 4 **Engine** stitches full `hb_*` set into GAM `cust_params`; applies floor bias; refreshes correlator.
- 5 **IMA → GAM**: fetches VAST; line item targeting `hb_pb` wins.
- 6 **GAM** returns wrapper with `%%PATTERN:hb_uuid%%` resolved.
- 7 **IMA → Prebid Cache**: fetches and plays the bidder creative; content resumes after.

FALLBACK

Prebid load failure, bidder timeout, or no-bid → engine still calls GAM with `hb_pb=0.00&hb_bidder=none`. Your house line item fills.

02 — Prerequisites

- **Google Ad Manager** account with video inventory; a **Master Ad Tag URL** ending with `correlator=`.
- **Limelight Digital** publisher ID + ad unit ID, plus the bidder host.
- **Domain allowlisted** by Limelight (1–3 business days). Without this, every auction returns no-bid.
- **HTTPS** publisher pages (mixed content blocks IMA + bidder calls).
- **Content video** URL (.mp4 / .m3u8) on your CDN.

#1 CAUSE OF ZERO FILL

The publisher domain not being on Limelight's allowlist. Verify *before* launch.

03 — Quick start

1 Generate the tag

Open the Tag Generator. Fill in version, GAM tag, Limelight publisher / ad-unit IDs, content URL. Copy the resulting `<script>`.

2 Set up GAM

Create the key-values, the VAST wrapper creative, and one line item per \$0.10 bucket. One-time setup — see section 05.

3 Embed

Paste the snippet where the player should render. The engine auto-creates the mount div.

4 Verify

Open with `?debug=true`. See section 08.

5 Monitor

GAM Reports → filter by `hb_bidder=limelightDigital` for impressions, eCPM, fill.

04 — Embed code reference

The **v2** tag uses `data-bidders` — a JSON array of every SSP that should bid in this auction. The Prebid bundle at `data-prebid-url` must include adapters for every bidder listed.

```
<script src="https://cdn.jsdelivr.net/gh/aryanvani-projects/bidding-player@v2.1.0/engine/p
layer.js"
  id="adtech-player-core"
  data-bidders='[
    {"bidder":"limelightDigital","params":{"host":"ads-jbi003...","publisherId":"649
658371","adUnitId":972556929,"adUnitType":"video"}},
    {"bidder":"appnexus","params":{"placementId":12345678}}
  ]'
  data-tag="https://pubads.g.doubleclick.net/gampad/ads?iu=/...&output=vast&correlat
or="
  data-video="https://your-cdn.example.com/content.mp4"
  data-prebid-url="https://cdn.jsdelivr.net/gh/aryanvani-projects/bidding-player@v2.
1.0/prebid/prebid-limelight-appnexus.js"
  data-timeout="1200" data-bias="0.10"
  data-autoplay="true" data-muted="true" data-fluid="true"
  async></script>
```

ATTRIBUTE	REQ	DEFAULT	NOTES
<code>src</code>	Yes	—	Versioned jsDelivr URL. Pin to a tag, never <code>@main</code> .
<code>id</code>	Yes	<code>adtech-</code> <code>player-core</code>	Must be exactly this value.
<code>async</code>	Yes	—	Required HTML attribute.
<code>data-bidders</code>	Yes*	—	v2 multi-bidder. JSON array of Prebid bid objects (<code>{bidder, params}</code>); each entry's <code>params</code> shape is per-SSP. *Optional if using the legacy 3-attr shortcut below.
<code>data-pub-id</code> / <code>-adunit-id</code> / <code>-host</code>	Legacy	—	v1 legacy shortcut. Engine builds a 1-element <code>limelightDigital</code> array from these. Ignored when <code>data-bidders</code> is set.
<code>data-tag</code>	Yes	—	GAM master ad tag URL; must end with <code>correlator=</code> .
<code>data-prebid-url</code>	Yes	—	Must include adapters for every bidder in <code>data-bidders</code> .
<code>data-video</code>	No	—	Content video (.mp4 / .m3u8).
<code>data-timeout</code>	No	<code>1200</code>	Bidder timeout (ms). Industry default 1500–2000.
<code>data-bias</code>	No	<code>0.10</code>	USD added to winning CPM before bucketing.
<code>data-autoplay</code> / <code>-muted</code>	No	<code>true</code> / <code>true</code>	Autoplay requires muted.
<code>data-fluid</code>	No	<code>true</code>	Responsive 16:9 vs fixed 640×480.
<code>data-loop</code> / <code>-preload</code> / <code>-vpaid</code>	No	see notes	<code>false</code> / <code>metadata</code> / <code>insecure</code> .
<code>data-div-id</code>	No	<code>comparos-</code> <code>video-</code> <code>placement</code>	Mount div ID. Auto-created if missing.
<code>data-cache</code>	No	(AppNexus public cache)	Prebid Cache server.

Onboarding a new SSP / Prebid bundle

The `prebid/` directory holds pre-built Prebid.js bundles. Each `data-prebid-url` picks one. To add a bundle with different adapters, trigger the **"Build Prebid bundle"** workflow in the Actions tab: supply the comma-separated module list and a filename, the workflow runs `gulp build` and commits the result on `main`. Full instructions in `prebid/README.md`.

THREE THINGS MUST STAY IN SYNC

1) `src` + `data-prebid-url` on the same version tag. 2) Every bidder in `data-bidders` must have its adapter in the Prebid bundle. 3) Bundle must be from the *same* `@vX.Y.Z` as the engine. The Tag Generator enforces all three.

05 — GAM setup

One-time setup: **key-values**, a **VAST wrapper creative**, and **price-bucket line items**.

1. Custom key-values

GAM → **Inventory** → **Key-values** → **New key-value**:

KEY	TYPE	VALUES
hb_pb	Predefined	0.00, 0.10, ..., 20.00 (one per \$0.10 step)
hb_bidder	Predefined	limelightDigital, none
hb_uuid	Free-form	(dynamic UUIDs)
hb_size	Predefined	640x360, 1280x720, 1920x1080
hb_format	Predefined	video
hb_adid	Free-form	(dynamic)
hb_source	Predefined	client, s2s

2. VAST wrapper creative

GAM → **Delivery** → **Creatives** → **New** → **Video** → **VAST 3.0 ad tag**. Paste:

```
https://prebid.adnxs.com/pbc/v1/cache?uuid=%%PATTERN:hb_uuid%%
```

Duration: **0:00** (wrapper resolves the real duration). Substitute your own cache URL if not using the AppNexus public cache.

3. Price-bucket line items

One line item per \$0.10 bucket (200 total for \$0–\$20). Use Prebid's **Line Item Generator** (docs.prebid.org/tools/line-item-generator.html) to bulk-create via CSV.

FIELD	VALUE
Name	Prebid HB \$2.50
Type	Price Priority
CPM	Match the bucket exactly
Targeting → key-values	hb_pb is one of 2.50
Creative	The VAST wrapper from step 2

EXACT STRING MATCH

Prebid emits "2.50" (two decimals). A line item targeting "2.5" will never serve.

4. House fallback + order

Add a low-priority line item targeting hb_pb=0.00 with your house creative or AdSense passback. Wrap all line items in one Order ("Prebid Header Bidding") for one-click pause/activate.

06 — Limelight onboarding

1. Apply for a Limelight account via your CSM.
2. Receive credentials: `publisherId` → `data-pub-id` ; `adUnitId` → `data-adunit-id` ; `host` → `data-host` .
3. Submit publisher domain(s) for allowlist approval (1–3 business days).
4. Verify bidding using section 08.

PRE-ALLOWLIST

You can test integration mechanics before allowlisting completes — the bidder returns no-bid and GAM's fallback fills.

07 — Consent management

Defaults are tuned for non-EU traffic. EU and US-state-privacy publishers override.

FRAMEWORK	DEFAULT
GDPR (TCF v2)	Static <code>gdprApplies: false</code> — non-EU-safe.
USP (CCPA)	Auto-detects <code>__uspapi</code> (100 ms timeout).
GPP	Off — module cancels auction without a GPP CMP.

EU — enable TCF v2

Place a registered CMP (OneTrust, Sourcepoint, Quantcast, Didomi...). Add this script *before* the engine tag:

```
<script>
window.pbjs = window.pbjs || { que: [] };
pbjs.que.push(function() {
  pbjs.setConfig({
    consentManagement: {
      gdpr: { cmpApi: 'iab', timeout: 8000, defaultGdprScope: true }
    }
  });
});
</script>
```

US state privacy — enable GPP

```
pbjs.setConfig({ consentManagement: { gpp: { cmpApi: 'iab', timeout: 3000 } } });
```

COMPLIANCE ≠ ENGINE CONFIG

The engine ships the technical hooks. Legal compliance depends on your CMP, privacy policy, and DPAs. Consult counsel.

08 — Testing & verification

Append `?debug=true` to your URL.

Console (filter for **STEP**)

- **STEP 1** Auction Initialized · Target: limelightDigital · Timeout: 1200ms
- **STEP 1.5** Requesting from Limelight · Player size: W×H
- **WINNER** <bidder> · Raw CPM, or **NO MARKET DEMAND**
- **STEP 6** Dispatching VAST Request

Network

REQUEST	STATUS
<code>cdn.jsdelivr.net/.../player.js</code> + <code>.../prebid.js</code>	200
<code>imasdk.googleapis.com/.../ima3.js</code>	200
POST <code>{host}/ortbhb</code> — inspect Payload	200
<code>prebid.adnxs.com/pbc/v1/cache</code> (winning bid only)	200
GET <code>pubads.g.doubleclick.net/gampad/ads</code> — decode <code>cust_params</code> , must contain <code>hb_pb</code> , <code>hb_bidder</code> , <code>hb_uuid</code>	200

Visual

- Pre-roll plays before content; Video.js controls (play / pause / mute / volume / progress / fullscreen) respond.

09 — Troubleshooting

No `POST /ortbhb` request in Network

- `pbjs.installedModules` must include `limelightDigitalBidAdapter`. If missing, fix `data-prebid-url`.
- Look for "Canceling auction as per consentManagement config" — see section 07.

`200` response but empty `seatbid`

- **Domain not allowlisted** by Limelight — the most common cause.
- Player rendered < 300×200 — bidders may reject small inventory.

Bid wins but no ad plays / GAM falls through

- `cust_params` must contain `hb_uuid`. Missing → engine < v1.2.0; upgrade.
- Line item targeting must match Prebid's string format exactly: `"2.50"` not `"2.5"`.
- VAST wrapper must use `%%PATTERN:hb_uuid%%`.

Request timeout after 1200ms

Increase `data-timeout` to `1500` – `2000` ms, or ask Limelight for a regional endpoint.

Autoplay doesn't start

Confirm `data-autoplay="true"` AND `data-muted="true"`. HTTPS required. iOS Safari has stricter rules with no engine-level workaround.

Mixed-content warnings

HTTPS page + `http://` resource (content / GAM / creative). Force everything to HTTPS.

10 — FAQ

Multiple players per page?

Not currently. Workaround: iframe per player.

Other Prebid bidders / multiple SSPs?

Yes — v2 supports multi-bidder via `data-bidders`. Ask your account manager which SSPs you need and they will generate a Prebid bundle containing those adapters.

If Prebid fails to load?

Engine catches it and calls GAM directly with `hb_pb=0.00&hb_bidder=none`. Your house line item fills.

Latency impact?

`async` script — zero impact on Time to Interactive. Adds the bidder timeout (default 1200 ms) before pre-roll start.

`@v2.1.0` vs `@main` ?

Tags are immutable (1-year CDN cache). `@main` = latest commit (~12 h cache). Pin tags for prod.

Auction analytics?

Impression-level reporting via GAM filtered by `hb_bidder`. Auction-level (bid latency / rate) requires a Prebid build with an analytics adapter — ask your account manager.

Bidding Player • v2.1.0 • Built on Prebid.js, video.js, Google IMA SDK

Engine: `cdn.jsdelivr.net/gh/aryanvani-projects/bidding-player@v2.1.0/engine/player.js`